Semantic Hash Storage

by Sven Nilsen, 2026

In this paper, I describe a path semantical theoretical perspective on Semantic Hash Storage that treats the path semantical qubit operator as fundamental, where `true` and `false` exist at the meta-level of language. From a Hegelian Dialectic perspective, this provides a way to look at the foundation of Path Semantics as a Synthesis between Logic and Semantic Hash Storage.

There has been a lot of philosophical debate and scientific research on semantics and storage systems. Sometimes semantics is treated separately from storage, sometimes they are viewed together. Humans have a complex history and complex social problems related to this topic.

For example, semantic triplets play a very important role in modern technology, although they are not very visibly used for a good reason. Semantic triplets are used to collect user data from social networks and process this data into a format that can be sold on the data market, legal or otherwise.

Therefore, when I write about this topic, it is kind of like opening a can of worms. Humans do not often spend much time thinking carefully about semantics and storage, but tend to race toward the earliest possible economic opportunity. A sentiment that, metaphorically speaking, hangs over the human species as a thunder cloud on the horizon, in the landscape of potential futures.

The question is ethically, why should we think about semantics and storage, if history shows it leads easily to possible futures where humans metaphorically tie themselves into knots, destroying themselves and their values in the process? Well, first of all, what is it about semantics and storage in particular, that makes it such destructive, despite that the very same research is used constructively, but also blended in with complex mechanisms of technology that both maintains and destroys human society? Nobody wakes up in the morning having these worries about logic.

One can use language bias to answer these questions. Logic is Platonic biased. Semantics and storage, when viewed together, is Seshatic biased. Since Seshatism is ethically ambiguous, this explains why there is a such complex social dynamics that both destroys and constructs society.

When it comes to logic, we humans have a hard time imagining why logic can be bad for society. Somehow, it is difficult to imagine this, despite that almost all modern computers use logic as fundament at a basic level of computing and modern computers are the machines that cause all the social problems around semantics and storage. Why are humans such biased in their thinking?

The research on Path Semantics indicate so far that human brains have something like an internalized Joker Calculus language, that controls how information gets organized. This explains the headaches people get when thinking about Joker Calculus, using computer networks as an analogy, as packet information that leaks over into the protocol layer of language, causing confusion and other mental symptoms. The reason humans feel less existential dread toward logic, compared to semantics and storage, is because Platonism and Seshatism are accompanied by mental phenomena, just when we think about them. Seshatism just feels ethically ambiguous, while Platonism just feels objective and tends to downplay ethical concerns. This is how our brains work.

Meditation exercises help people to bring these language biases under control. With other words, there is a way our brains work, that make us pay more attention toward negative aspects of semantics and storage, but there is no reason to think that way when looking at the bigger picture and taking all the evidence into account. This brings us to a starting point for Hegelian Dialectics.

Hegelian Dialectics is a misnomer, because most people think about it as Thesis, Antithesis and Synthesis and believe that Hegel was the one that thought about this first. That version of Dialectics came from Fichte, an earlier philosopher, contemporary with Hegel and one of his mentors.

Well, depends on what you mean by mentor. Human social relationships are complex and some might even call Fichte a co-conspirator to Hegel, in the sense that they were not really doing philosophy as such, but scheming and plotting, about the grander scheme of things. While history did not turn out the way that Fichte and Hegel with others in their social circles became supervillains, it might have turned out that way, or one might question, whether there is at least a supervillain-ish aura there. In modern equivalent, but much more as pathetic losers, one can think about some kind of people that are working for CIA, that are plotting to take over the world, while also unable to escape the comfort and routine of having a steady job and income. A human metaphorical toilet drain, that might as well have sucked up Fichte and Hegel before they managed to produce works that made the world go mad, because their careers did not allow such laziness.

Most likely, your ideas of what Hegel was actually thinking about, can be written down on a post-it note and thrown into a trash can. If you forgot what was on this post-it note, which is everything so far, neither you nor the world would be in a worse state. You have not read Hegel (most of you) and even if you did, there is so little that you would learn from it that instead, you would go with the conformist view. But, the sad truth, is that there is still so far between what you think the problem is, with that level of ignorance, and what the actual problem: Hegel is a philosophical satirist.

The problem with Hegel Dialectics is that even if people could wake up from their slumber and just read him and think through all the stuff he wrote, it would not help much, because Hegel is not a serious philosopher. He is an actual philosopher, living under a system of oppression. That is something far worse. Because, in order to understand how actual philosophers are thinking, you would need to learn actual history and not the cartoon version for kids you learned in school. If you come from a rich family in addition, then your biases already make you screwed, because you might as well live on the Moon as far as your imaginary mental world resembles the planet Earth. My dry humor put aside, in Hegel's philosophy, the intellectual world of the elite is constrained and only the slave philosopher can really put things together. He digs up a Seshatic layer that is buried within Christianity, like all the other philosophers who developed a secret admiration of Spinoza, that not before Einstein dared to venture out of the closet enough, to express this in form of poems.

There are tons of books written about Hegel, but in the end, all what people get out of his works, is what they create in their own imagination. And it is all Darwin's fault. You see, the year Hegel died, Darwin went on a journey, sailing around the Earth, to gather scientific evidence about evolution. As much people want to believe that the ideas of evolution and natural selection started with Darwin, is because they do not study actual history (because who needs to know what actual philosophers are thinking about these days, not the serious ones they push on you in school, if you are lucky enough to reach the level of social privilege to receive a such intellectual abuse). However, it is not incorrect to say that it is still Darwin's fault. Not the Charles, but Erasmus, his grandfather. These ideas of evolution, occasionally written in the form of erotic poems about plants with scientific footnotes, that were enabled to circulate in Great Britain that did not have a central institution for censorship, to great degree of scandal were read by women (!), were smuggled on boats and reprinted in France, near the Swiss border, where the Large Hadron Collider is placed today (for the same reasons). In the 18th and 19th century, there was an intellectual underground movement, that due to the trends of society losing interest in metaphors, is now under the ground.

Naturally, or shall I say by natural selection, the Christian academics on the Continental Europe, who sucked up to various monarchs, were eagerly reading these banned books while struggling to invent new ways to hide in their works that they had even been nourishing themselves intellectually.

The word “empiricism” comes from a Roman school of medicine, The Empiricists. Your cause of misery and ignorance comes from a competing Roman school of medicine, The Dogmatists. Yes, I know you thought Christianity was a form of religion. That is what they taught you, right? Christianity was a form of practicing medicine, that used Galen’s works, you know, all the banned Pagan practices of medicine, to create a monopoly on healthcare. That is why they burned women on stakes, who tried to start up their own private business based on Pagan practices of medicine. It was literally the same knowledge of medicine, but protected by a monopoly that pretended it did not use this knowledge, against all other entrepreneurs. OK? So, basically, your idea of Christianity is in-dated (the same as out-dated, but backwards in time) around the early to mid 19th century. Or, the year 1831, to be precise. The year Hegel died and when Charles Darwin went on an epic journey.

There is almost nothing you know about Hegel that was not invented sometime later than 1831. If you graduated from Harvard or Yale in addition, with a heavy mountain of debt from all these courses in philosophy, you now know how modern scams are performed. Move on with your life. As I said before, to understand Hegel, one of the first principles is that actual philosophy is not serious philosophy and only those who care about actual philosophy read actual history, or otherwise you have no chance to understand what actual philosophers are thinking about today. This sounds like a joke. The actual joke here is on people who pay to learn a fake version of history and serious philosophy, which is about as relevant as the time it takes for ink to dry. If you come from a privileged background, to a such degree that you put yourself into debt, then you might see me as a projected target for your own inner insecurity, but remember, I am not the cause your trouble. I am just trying to explain Hegel’s philosophy to people who know nothing, learned nothing and all they “know” fits on a single post-it note. In addition, I do this without taking paid for it. In comparison, what good have you done for society? Maybe you should think about that, instead of attacking me.

Hopefully, I have now managed to get rid of the readers who are those kind of people that tend to seek the earliest economic opportunity to exploit research on semantics and storage. This is how I deal with these people, by offending them, because otherwise they will not leave voluntarily. Now, you and I are on the same page, where I will follow by seemingly dismissing Hegelian Dialectics and use Fichte’s Dialectics instead: Thesis, Antithesis and Synthesis. With other words, what most people think is Hegelian Dialectics, but is not. I did not lie to you. Society did. Not my mistake.

For programmers, the idea of Thesis, Antithesis and Synthesis sounds like something designed for people who have no idea of how computers work. It is about the dumbest thing you can make. First, it does not work. Second, it is inconsistent with how logic works in general. It sounds precisely as something people would say, to feel smart about themselves and impress others like them. They do not realize how hard it is to impress a computer.

To try think about Thesis, Antithesis and Synthesis from a perspective of logic, will fail. Therefore, I will explain it to programmers in this way: If A is Thesis and B is Antithesis, then you can put them together in one expression and calculate a hash. From the hash, you look up C, the Synthesis, in a hash table. Basically, that is how it works.

Now, you programmers can leave. Please.

OK? There are some of you that are programmers left, because you realize that hash is implemented internally in hash tables, not externally. Either you thought I misunderstood how hash tables work, and you left believing it for the rest of your life, or you stayed, thinking I had a deeper insight.

What I am talking about here, is not implementations of hash tables specifically, but the idea of hashing in general. Why do we use hashing in programming? The most common perspective is to get `O(1)` performance. However, here we need to take a step back and think more carefully.

Why would somebody use hashing, even though it has nothing to do with performance in principle? One property of hashing, which is frequently used in cryptology, is that hashing obscures the information in the original expression. It functions as some kind of safety mechanism. However, it is also important to notice that hashing is most commonly used on infinite sets of data, such as strings or bits. We are talking about very large complexity here. While the actual information that we feed to a hashing algorithm usually terminates after finite time, there is in principle no bounds on that finite information, making it one sample of an infinite space of complexity.

As a programmer, you already know this from experience. However, you might have never actually thought about it. That knowledge exists in tacit form, something that you recognize when other people point it out. Sometimes, it might even feel like, that you are not supposed to talk about or express such knowledge explicitly. It kind of ruins the psychological flow of your life's narrative.

This sentiment is part of the reason, why programmers reject philosophy, because they recognize, that all this debate would not have been needed in the first place, if these serious philosophers just learned how to use computers properly instead. However, remember earlier, according to Hegel, there is a difference between serious and actual philosophy. An actual philosopher today can actually talk to programmers, not in terms of abstract concepts, but by explaining things in terms of programming terminology. Therefore, I will explicitly point out what you already know.

Logically speaking, when an infinite set is reduced to a finite set through hashing, there will be some hash collisions. This seems self-evident, but it might be tricky to actually prove this formally yourself. There is a lot of ambiguity here, about what one means with infinite set or finite sets, or whether they should be thought of as one class of all sets, or as two separate classes. Any formal proof would need to depend on the precise definition you use for sets. Now, the question is what logical language you should use. First Order Logic and Zermelo-Fraenkel Set Theory? One problem with this approach, is the sets in Zermelo-Fraenkel Set Theory are not actual sets, but trees without symmetries, according to Vladimir Voedvodsky. In any case, when you try to think about how to prove formally that there will be hash collisions when reducing an infinite set to a finite set, you run up against problems with the formal language. If the language is too formal, then the proof might apply only in a limited context that requires additional work to show that it holds for some specific application. If the language is too informal, then we are no longer sure what kind of notion of truth or sense that we are using. Then, you realize: There is simply no perfect way to describe this idea.

The idea that hashing leads to philosophical ambiguity, is much more important, than the specific properties of hashing. In a sense, this leads you toward the position, where you can see how a serious philosopher would demand a formal proof (if they could understand how formal proofs worked, not their fantasy version of absolute control, they would, but they do not), whereas an actual philosopher would say something like "welcome to the club!" and give you pat on the back.

Actual philosophers today understand how hashing works. They also know why it leads to philosophical ambiguity. Plus, they also know this insight is much more important than the details.

Actual philosophers read actual history, to learn what actual philosophers are thinking about today. These people know how this stuff works and can put ideas that evolve in society over time in context of technological progress, even the complex relationship humans have with astronomy.

If you do not study actual history, then you can not do actual philosophy. That was Hegel's idea, his notion of Dialectics, which holds pretty good even today. For all Hegel's shortcomings otherwise, this is pretty impressive, when you think about the fact that he died a whole century before the first binary computer was put into use. Not even including the period of 50 following years when programming were done by women and put men on the Moon, as one minor accomplishment.

When men discovered that programming was fun, programming had already changed the world. That is why, computer scientists do not learn computer history today. They have no need to understand computer history, more than Christians today have need to learn about Early Christianity, when women were leading and seeking to use knowledge to make social progress.

Have you ever thought, as a programmer, that what you know about the history of your own field can be written down on a post-it note and thrown into the trash can, without making your own life or the rest of the world anymore worse than it already is? You are a useless person.

Yeah, you got angry? Sure. You deserve no respect from actual philosophers.

I just pointed out to you, how hashing of an infinite set into a finite set can explain philosophical ambiguity. Yet, while you are still are angry at me for calling you useless, you have not realized what is so important about this idea. Why would anyone think about it as important, at all?

Can you put your anger on pause, just for a while? Let us think about it, you and me. Together, maybe we can figure it out. I know it is an illusion, because I am writing this before you read it, so in principle you can not think it through. Instead, it is more like, I am guiding your intuition forward, to see the next natural steps to take. However, since your Ego is such a bottled up angry and insecure-isolating-feeling part of your psychology, it helps to pretend that you are thinking it through with me, so you will allow me to continue. I am just playing with you. Anyway.

You managed to get all this way through this paper, so you will not give up now. Not when you are so close to figure out why I would write a paper about Semantic Hash Storage.

Seems that you are absolutely hopeless as an individual. I like that.

When we enter the domain of philosophical ambiguity, the most common technique is simply to separate things apart, so instead of having one hashing mechanism like this:

expr → expr? (we look up an expression in a hash table that uses hashing internally)

This gets divided into two parts:

expr → hash hash → expr?

The `?` means that returning an expression is optional. There might be no such entry in the table.

How most hash tables are implemented, is that they take into account the risk of hash collision. Instead of storing one entry per hash, they store a list, such that if there is a collision one can fall back on equality check. Now, if you are a good programmer, then you designed your application such that the hash is not too expensive, but also, that it has very low risk of hash collision.

However, in the real world, people do not have time to think about stuff like hashing, so in practice there is almost nobody, except those who work in cryptography, that need to think about it. They do not need perfect performance, just good enough.

Now, what if you were seeking perfect performance? What then?

You would not need a list per entry. Instead, you split up hash table in two functions like above, because it provides better CPU cache locality. It is almost as if just by assuming some von Neumann architecture, which feels like a natural assumption, this scheme is a natural next step.

The thing about Hegel's philosophy, is that there is learning curve, from the range of IQ where you live in blissful happiness, to the sad range of IQ where you want to feel smart but can not, to where you are like most people who strive to find a steady income while not be willing to cut back on consumption, to most smart people who learn from serious philosophers, to even smarter people who program computers, to even smarter people that realize that Hegel is a satirist, to the even smarter people who understand that Hegel is kind of like a very smart programmer.

When I was saying earlier that I was going to use Hegelian Dialectics, I was not lying. Just because I said, society has been lying to you and what you think of as Hegelian Dialectics, is actually Fichte's version of Dialectics, does not mean that I will be using Fichte's version from now on. So far, Fichte's version was useful to show how we move from Post-Kantian philosophy to programming. Once you realize this is about hashing two things, by putting them into one expression, you can kind of discard Fichte's philosophy, for now.

There is something about hashing, which is peculiar for computers with finite memory and that operate within a limited range of time. In practice, one can design algorithms such that there will be no expected hash collision. Not even once. In the real world.

This is an amazing property from a philosophical perspective, because it is kind of like cheating. We want to feel like that we are living in some perfect Platonic world, and there are some ways to do justify some of that feeling, by exploiting tricks that makes something finite and mortal, for its time of brief existence, creating an illusion of Platonism. Perfect hashing is one of them.

I will let you live for a moment with this feeling, before I pop your bubble later on. However, just right now, hold onto that feeling. It is good, right? Just for once, something can actually work.

The issue here is that there are two functions, one with type `expr → hash` and one with the type `hash → expr?`. Since they are separated and there is no way to use lists here, you kind of have to think like a very smart programmer and bet that the hash is going to behave perfectly. Now, the chances is that you might feel a little bad about it, because you know it is not 100% perfect. It will just never fail. There is a difference. Still, if we can assume that you were a smart programmer, which you are not, then it feels good to think about the hash just working perfectly in practice.

Did you notice that during these steps, the way that you were thinking, changed?

You get a good feeling about the hash working perfectly, because you are not a smart programmer. A smart programmer would just be bored. It is like, when a poor person finds a bag of money. First, they get happy and spend it all, but later on they start worrying the people who owned that money will come after them. They have not realized yet, that people with money do not care about money, they just like to hunt down anyone that did anything remotely resembling to hurting their feelings, because their mum did not make waffles with them as kids.

No, people are not evil because they have to. They are evil, because they like how it feels. There is a difference. Hegel uses the master and the slave as an example. The master likes how they treat the slave, despite that they would never appreciate being treated as a slave themselves. This is the principle of asymmetry: Power feels good to people, precisely because other people do not have it. When they meet actual intellectuals, their brain just crashes and burn. It gets fried. Like waffles.

A "waffle" is a person who is wasting their life, so that waffles are more interesting to think about.

You see, people think I am offending them, because they have power and I have not. The real reason I am offending them, is because they are wasting their lives. The waffle is just a waffle. It is real.

To somebody who grew up without their mum making waffles with them as kids, they can not think about the “waffle” as something real, because it is not in their own experience. They think about it in an imaginary way, such that, they have to grasp fully what the waffle means with their own mind.

That is kind of like, how people think about programming before they have done any actual programming. After you have programming experience, you stop treating it as imaginary. It is real.

Programming changes how you see the world. After having programming experience, you never again see the world in the same way you did before. There are still infinite possible futures, just like before you learned programming. The difference is, that in all the infinite possible futures, unless you lose your memory, the experience of programming will influence how you see the world in every one of them.

This is an integrated part of Hegel’s philosophy. You are now starting to understand how Hegelian Dialectics works. Not the cartoon version that is Fichte’s philosophy dumbed down and repackaged, but the real Hegelian Dialectics. Just in the simple example of hashing, our way of thinking changes, and depends, on every stage of the process thinking it through, influenced by our personal experiences. You might feel good at some stage, because you are not used to succeed at doing stuff.

Do you now understand why Hegel is a satirist?

Actual philosophers are so far beyond conventional thinking, that just because Hegel was a fanboy of Napoleon and Charles Darwin went on a journey around the Earth after his death, your ideas about Hegel are based on all that happened after 1831, after Hegel was dead, instead of actually understanding what Hegel was doing. You are nowhere close to the intellect where Hegel was, nor will you probably ever be. It is not about IQ, but your extreme lack of sense of humor.

With other words, you are talking to a dead person. You are not talking to Hegel, addressing him, the way he was while being alive. A douchebag. A man who built a whole philosophy on the brilliant idea of marketing it in a system for the living room alongside silhouettes, a precursor to photography. Silhouettes on the walls, Hegel’s works on the shelf. That is a perspective that you will never understand, because you live in a world with photography and did not experience how significant silhouettes played in social change of people’s self-perception at the time. That people could be remembered. That people have a history. You know, because up until that moment, the only people who were remembered were the rich who could afford paintings.

Your extreme lack of sense of humor, depraves you of the idea that Hegel was not only bridging the gap from you as a programmer to understanding how smart programmers are thinking about hashing, but that his starting point is from a time where ordinary people are not even remembered after they are gone. You? Against Hegel the satirist? No, Hegel became an easy target, precisely because of Darwin’s outside influence on philosophy as a field. Everyone wants an easy target, but they do not dare to open up history books and study them. Hegel’s philosophy shaped a whole world view, where people started seeing history as not about the rich, but about themselves. How they looked. Which school they went to. Where they traveled.

Silhouettes were just a starting point. If you looked too fat, the artist could trim your belly a bit. Photography was the next step. Freckles, wrinkles, bad lighting, or removing an entire person in case of dictatorship, required delicate manipulation of information to present the past as intended, while hiding all the ugliness, the errors and bad habits under the rug. Photography was kind of like hashing. It is harder to fake one. So, the only way for society to progress, was by increasing the amount of information that got manipulated intelligently per second. At the end of this process, there is this idea of the perfect hash: Something that nobody can manipulate. Absolute Knowledge.

Hegel’s philosophy is about thinking about history and what it means. Hegel never knew about hashing. However, he got a vague idea of how such things might relate to history. What the things people would look for in such contexts. The problem is that today, even if you examine from Harvard, then the education might be not worth the printed diploma, the way some of those people are waffling their way through life. They still look for that imaginary thing. The waffle they seek.

Sometimes, people are so stupid that they look for waffles they can easily grasp with their minds, instead of just enjoying real waffles. A lot of people are evil, not mostly because they do bad things, because they are just thinking about doing bad things and believe that they are building a system of information processing intended to do evil, while lying to the rest of society that they are “just” going to destroy civilization and rebuild it from ashes. Because you know, it feels better that way.

When such people simply get arrested and end up in jail instead, they always get surprised. If you read any actual history, then one thing you notice is that such people get arrested every single day.

Trying to explain Hegel’s philosophy to people today, is almost impossible, because even the people that graduate from Harvard these days are seeking the earliest moment of economic opportunity while playing wanna-be supervillains. Hegel, on the contrary, would make a fine supervillain if he put some effort into it. The number of people who died due to Hegel’s works is estimated in hundreds of millions, but they mostly happened because of dumb people trying to revolutionize philosophy since they were trying to make something as revolutionary as Darwin’s evolution.

Basically, that is the mental world you live in. It is all crap.

Maybe, if you could time travel, going back to Hegel before Darwin, then you could start work on your understanding what relationships people have to history, all the while feeling the aura of some conspiracy going on, in the air, a haunting feeling that somebody are watching you. Because, they were. The monopoly on healthcare, that got called Christianity, because people needed some version of history they could believe in, was going through the works of contemporary philosophers, line by line, to see whether there was any trace of rebel ideas. People cutting off the heads of a few monarchs was acceptable, as long they did not start up a competitor of medicine, where the big money was at the time, long before fossil fuel.

This puts history in a different perspective, right? You are not an objective observer of history. The brain seeks out stories that it likes, what feels good, right there and then. Remember the good feeling of the hash working perfectly? I am going to pop that balloon now, as promised.

The functions `expr → hash` and `hash → expr?` allow side effects. So, instead of a hash, you pick a number type with a range that holds for all practical applications. In each of these two functions, you simply use an ordinary hash table implementation that falls back to equality checks on collision. The number you use is incremented over time, so that the only collision that can happen is when this number is incremented beyond the range. Now, it works perfectly, in all possible worlds, within the limited time that the applications are used. What a bummer.

You thought I was only joking, when I said you are not a smart programmer? I admit, I was misleading you. The idea was that, making you think the point was about making the hashing work perfectly. Not the way I did, which fixed the problem perfectly, even the hashing is internally imperfect. Unless you could figure this out yourself, I just proved that you are not, in fact, a smart programmer. I never promised that the solution would be impressive. It is you thinking that way, because you live in an imaginary world where you have ideas about how smart programmers are like. That is why you also fail to comprehend what actual philosophers are like. Actual philosophers, are not serious philosophers, who try to impress. Actual philosophers get stuff done.

Now, I also said earlier, that you deserve no respect from actual philosophers.

Respect is something that usually happens in a limited social context, where the meaning is that the intention of that social context is serious. However, most people never realize, that there is a way, in some social contexts “no respect” is how you actually treat people well. By satire. By making jokes.

I am trying to teach you something about Hegel’s philosophy. However, as an individual person, you are a hopeless case. Just the fact that you are still reading this paper, proves my point.

How long do you think you can keep on, letting me lure you into this deep rabbit hole? Because, this was just the beginning. From now on, I am going stop holding your hand with silk gloves.

OK, time to do some actual philosophy.

The whole reason that we are talking about hashing here, is because hashing enables a form of information organization that computer scientists call “unstructured data”. It means, data that is more freely organized. Think about using a chalkboard as an example. On the chalkboard, you can write and draw whatever you want, while still having an overview of what you are thinking about. Somehow, the things you write at home on chalkboards will stay up for so long time that you can not get off the marks. So, you go back to paper, because you know, paper can just be thrown out or put in plastic folders. However, after some time, you go back to chalkboards again because paper sucks at getting just the right feeling you need to be creative. The idea of “unstructured data” basically means all the mess that you create, while pretending to work. If you want to get stuff done, then you have to make a list and the type of the list must be an array of strings. That is an example of structured data. We know you can not mess it up, because the type is `[str]`, right?

Hashing is useful, because nobody knows what it means just by looking at it. In Semiotics, we call a such sign for a symbol. Symbols are signs that you can not figure out by observation alone. You have to learn something more, or receive instructions, or go to the library that is perhaps not the biggest library you think you want, but a smaller one with a better librarian, that chose the books on the shelves you never thought you needed to read. Usually somewhere where they organize yearly festivals for philosophers. That kind of limits it down, to one small town in Norway. Anyway.

The point is, since nobody really understands Hegel’s philosophy, and nobody understands hashing just by looking at it, those two things have in common that people do not understand them. So, people make up whatever they want these things to mean. That is what programmers do.

When you buy a computer, it starts out with a long list of bits, basically `[bool]`. It is even simpler and more structured than `[str]`. You can not mess it up, right? Still, all evidence so far, points in the direction, that you will manage to mess it up just fine. Furthermore, you will always be able to mess up any new computer you buy, no matter how advanced it is. All you need is `[bool]` to mess up.

Something is not right here. If you mess up things less with `[str]`, but more with `[bool]`, then is there really an order of preference with respect to messing up stuff? Is hashing really any better or worse, than any other technique?

Maybe not, but if you are smart enough to come up this hashing scheme of two functions `expr → hash` and `hash → expr?`, you might feel that this is so brilliant, that you never have to invent another data structure for your problems. You can just keep using the same idea, regardless of much mess you make out of it. Here, the basic idea is that, instead of adding additional structure beyond these two functions, we keep going inventing new tricks with side effects, that allow us to solve the problems and keep using the same way of organizing information.

It turns out that the only difference between a modern internet browser and this hashing scheme, is an impractical user interface. Despite all efforts, nobody have really managed to make a good one. So, maybe, getting rid of the entire user interface is better than having any? It might not look as good to most people, but it might actually be good. That is the basic idea, anyway.

OK, so now we got rid of the user interface, an internet browser is basically the same thing as this hashing scheme. What is the big deal? What is the next natural step we can take?

Well, think about all the stuff people use the web for. No, not that stuff. The legal stuff! As the case is for this paper, we will only concern us about things that are legal and useful. However, somebody needs to make the laws, right? Tor is a guy that lives down the street and is responsible for making up all the legal stuff. How does Tor put it in? I mean, how does the legal stuff gets on the web?

We need a particular side effect to put new things in. There is no way we can use `hash → expr?` to do that, because we need a hash and this would tangle up things that already exist. So, the only way to put new things in, is by using `expr → hash`. Now, to separate this process from the others, we need to say that we want to create something new. An expression “create <expr>” where the inner expression is the expression we want to create. The `expr → hash` function can return the hash of the new expression created, so we can look it up later. Brilliant!

Basically, whenever you got a new problem, you modify the expression type, so that it includes the new use case you need. You can keep doing this, I promise it will be messy though, but in principle, you can just do this over and over and never change the basic design, ever. You just end up with an extremely complex expression type and side effects that are very weird. Now, you have reached Absolute Knowing and in the process, turned your program into the only incomprehensible object.

This is what philosophers do: They invent a new type that they want to use to solve all their problems, including those problems they have not thought about yet, so that nothing ever needs to be invented again. The Ancient Greek philosophers invented `[str]` and produced the language that is the reason English is most used in the world, because English has fewer words and most of them are borrowed from Ancient Greek. English is wanna-be Ancient Greek but you do not have to think (Bacchus forbid!) and you do not need a sense of humor to use it. Later on, Leibniz invented `[bool]` and paved the way for the thousands of programming languages, that refuse to use 0s and 1s directly. A century after Leibniz, Hegel invented hashing before hashing existed, without anyone knowing about hashing and even without knowing it himself.

Hegel’s philosophy is a perfect example of hashing: It is completely obscure and incomprehensible, but people can use it to make up whatever they want, even if it is just by copying ideas from Fichte and claim they were invented by Hegel.

Great! So now we live in world where we neither understand Hegel nor Fichte. Problem solved.

Now, what is `true` and `false` in sense of Hegel and hashing, not in the sense of Leibniz? Leibniz would say that `true` is 1 and `false` is 0. These are just labels we use on states of information. However, when we think about `true` and `false` in constructive logic, we can not say that some proposition is either true or false. So, it is not just labels we use on states of information.

When we think about constructive logic, `true` and `false` carry meaning in relation to the rest of the language we are using. From `true`, we can only prove statements that are already true. From `false`, we can prove anything. So, if we glimpse a bit on the big picture, then we can say that `true` is kind of like empty and `false` is kind of like full. In this sense, related to hashing, there would be like `true` is when internet is empty and `false` when it has every possible combination of bits.

In either case, the internet would not be useful if it was empty nor when it has all possible combinations. There is a such thing, as people making cat videos because it is fun to watch, an emotional response you do not get by looking at a video where pixels change colors randomly. In the vast amounts of possible videos, most of them look indistinguishable to a human than something you can output from a random number generator. Cat videos are fun because they are narrowed down to a space where they are hard to make. People like them particularly because they unconsciously perceive cat videos as not made by anyone directly, but they discover something new, both because it will be random content and it is about one thing they love: Cats.

With other words, people like the internet because it is unstructured. It is random, but not so random that it gets boring. Think for example, of all the websites you could visit and read about more interesting stuff than this paper. Why would somebody even read a paper with the title “Semantic Hash Storage”? That sounds to me like the most boring paper, ever.

The entire reason why somebody bothers with Hegel’s philosophy is because it is random and still manages to trigger these small levers in people’s heads, that make them see something imaginary that they feel is relevant to their lives. Hegel was an expert in making people feel that they were important, specially the students who listened to him in the class room. Him and them, all males, where sharing this historical moment, together. Yeah, definitely not a supervillain. Just a satirist.

Something like the internet is needed to make people feel special. On the internet, people can hand over hundreds of billions to a billionaire that promises to wipe out the human race, while the people who built the internet do not receive any pay, but has to do this work in their spare time. You can find the same pattern across the board: There are always a few people who do most of the work, while most people seek out ways to make themselves feel more important, respected, acknowledged etc. by posting cat videos and worse while pretending to work.

Yeah, sure. You are definitely much smarter than Hegel. The philosopher that came up with the basic principle on which stuff like the internet is based on, is surely a lot less smarter than somebody, who managed to not learn a single thing their entire stay on Harvard, and came out with only a piece of paper and a mountain of debt. You know, Hegel had to work without pay as a teacher, because it was expected they got payed through giving private lectures in addition, before he became a professor. Call me back when you have given a lecture at Harvard.

My point is, `true` and `false` in sense of Hegel got nothing to do with labels you use on states of information, but about empty and full in terms of network knowledge. So, `true` and `false` in this sense exist only at the meta-level of language.

Wait a minute!

What I am saying, now that I think about it (for your sake, I will pretend I did not see this coming), that if you have a hashing mechanism, then in principle, you can create a system that is even more fundamental than `true` and `false`, because these ideas are emerging from the meta-level of a lower layer. The next natural step is to ask: What is the analogue of a such hashing mechanism in logic?

Surprise! Surprise! This mechanism is the path semantical qubit operator.

What? I hear nobody clapping? Not one “yay!” out there! Where are everybody?

As I said before, doing actual philosophy is not the same thing as doing serious philosophy. Actual philosophy is just getting stuff done. Nobody gives you a medal for this. Otherwise, they would need to give you a medal every time it happens, which would be unbearable for society.

Serious philosophy is about prestige. It is about, making that one comment that people can really understand and feel smart about themselves. Takes years of practice to become a serious philosopher, far more demanding, than just decades of experience it takes to become an actual philosopher. When you have become an actual philosopher, you are already so old, that people no longer pay any attention to you. Which is just fine, because you can work on whatever you want.

Using Semantic Hash Storage as a basis, one can look at the foundation of Path Semantics and think about it in a new way: There is Logic in Path Semantics, but there is also a mechanism like Semantic Hash Storage. So, if Semantic Hash Storage is Thesis and Logic is Antithesis, then Path Semantics is the Synthesis. In Path Semantics, there are explicit symbols for `true` and `false`, which one uses to understand what these ideas mean in sense of Hegel's philosophy and Semantic Hash Storage. When there are no explicit symbols for these ideas, but only semantics, it becomes much harder to recognize. One of the uses of Path Semantics is to reason explicitly, so that one does not need to do all this extra work for each particular domain of human knowledge.

When you want to create a practical implementation of Semantic Hash Storage, you will find that there are all sorts of edge cases, where the code has to be just right to make it fully functional. Somehow, humans like the big ideas about Semantic Hash Storage than diving into the details of it. Almost as if, people have the ability to distinguish a philosophical kind of mindset, separated from putting in the actual work. Like always in the AdvancedResearch community, it is much easier to just provide a functional reference implementation, so you can experiment with it as a programmer. This reference implementation has not been built yet.

Can you really build an entire internet browser based on this design? Yes you can! However, how would you know, if you never tried it yourself first? Maybe you should think about building stuff instead of worrying about people calling you a "waffle" for wasting your life, or something? However, please do not make another one of these databases. There are hundreds of them out there already. Most of them are not much better than using a text file. Who am I kidding? The people who make it this far in the paper are the kind of people who will just build what they want, no matter if it takes thousands of hours and nobody will use what they make. They do it for fun.

THE END

CREDITS: Starring Hegel and Fichte as themselves.

No animals were harmed when producing this paper.
Except my cat, who is pestering me to go buy one of those meat boxes she really likes.

(References? Hmm… exercise to the reader? OK, maybe later.)